

Lorenzo Marinelli

Curriculum Vitae

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ABOUT ME

Research profile MSc student in Computer Science at Sapienza University of Rome, with a BSc in Applied Computer Science and Artificial Intelligence awarded with highest honours. Research interests include graph representation learning, bioinformatics, computer vision, adversarial machine learning, and the robustness and reliability of machine-learning models. Experience includes end-to-end experimental pipelines, quantitative evaluation, reproducible software development, and deployment-oriented prototyping.

EDUCATION AND TRAINING

- 2025-present

MSc in Computer Science
Sapienza University of Rome
Graduate studies in computer science with particular interest in deep learning, graph-based methods, robust evaluation, and research software engineering.
- 2022-2025

BSc in Applied Computer Science and Artificial Intelligence
Sapienza University of Rome
Degree awarded 110/110 cum laude. English-taught programme, class L-31. Bachelor thesis on graph-neural-network link prediction in protein-protein interaction networks.
- 2017-2022

Scientific High School Diploma
Liceo Scientifico Statale "A. Romita", Campobasso
Diploma awarded 100/100 cum laude.

RESEARCH EXPERIENCE

- 2024-2025

Bachelor's Thesis: Protein Interaction Link Prediction
Unveiling Protein Connections through Graph Neural Networks and ProtT5 Embeddings
 - Developed an end-to-end pipeline for protein-protein interaction prediction using STRING v12 data, ProtT5 sequence embeddings, and graph neural networks.
 - Combined TAGConv, multi-head TransformerConv, and GIN layers with a pairwise decoder; implemented negative sampling, class weighting, edge-dropout augmentation, and reproducible evaluation.
 - Achieved 0.96 AUROC, recovered 81% of withheld positive interactions, and obtained $P@500 = 1.00$; examined threshold selection, class imbalance, and topology-driven bias.
 - Conducted a NOTCH2 case study to rank biologically plausible interaction candidates and documented methodological limitations and robustness-oriented future work.

SELECTED PROJECTS

- Adversarial ML

NoWhisper: Adversarial Robustness of Speech Recognition
 - Co-developed a framework for studying untargeted adversarial perturbations against Whisper, with the objective of degrading automatic transcription while preserving the intelligibility of the underlying speech.
 - Implemented sample-specific white-box attacks and training for a reusable neural perturbation generator, including amplitude constraints and robustness-oriented audio transformations.
 - Built reproducible LibriSpeech preparation, evaluation, and reporting workflows using word and character error rates, together with command-line tools and an interactive interface for attack analysis.
- Reinforcement learning

RL-Nav: Language-Guided Navigation under Partial Observability
 - Developed a stochastic PyBullet navigation environment containing obstacles and hazardous regions, and trained a recurrent PPO agent from partial observations.
 - Integrated an LLM-based oracle that returns constrained TURN/WALK advice, with canonical parsing, query cooldowns, failure handling, and a geometric planner used as a reference policy.
 - Evaluated oracle agreement, navigation success, and the propagation of local instruction errors into long-horizon behaviour, highlighting the gap between plausible language advice and reliable control.

Computer vision **UAV Anomaly Detection and Localization - SASA Flight Team**

- Developed an unsupervised anomaly-detection pipeline for high-resolution aerial imagery using autoencoder and variational-autoencoder reconstruction errors.
- Converted dense error maps into stable detections through denoising, threshold selection, morphological post-processing, connected-component analysis, and bounding-box extraction.
- Structured the pipeline for autonomous-aircraft deployment and downstream classification, with emphasis on noisy inputs, edge cases, modular implementation, and quantitative validation.

DIGITAL AND TECHNICAL SKILLS

Programming	Python; R; JavaScript; SQL fundamentals.
Machine learning	PyTorch; PyTorch Lightning; scikit-learn; adversarial evaluation; model training, validation, threshold analysis, ablation studies, and debugging.
Graphs and networks	PyTorch Geometric; NetworkX; Gephi; graph embeddings; graph neural networks; link prediction; network analysis.
Scientific computing	NumPy; pandas; Matplotlib; data cleaning; feature engineering; statistical analysis; reproducible notebooks.
Computer vision	OpenCV; torchvision; Pillow; autoencoders and VAEs; anomaly maps; segmentation post-processing; localization.
Speech and audio	Whisper; LibriSpeech; adversarial audio; WER/CER evaluation; constrained perturbation generation.
Bioinformatics	STRING; ProtT5 embeddings; Enrichr; WGCNA; SWIMMeR; SaveRunner; interactome and enrichment workflows.
Software engineering	Git; Docker; modular Python; logging and experiment tracking; artifact management; REST APIs; OpenAPI.
Synthetic data	Blender; domain randomization; annotation generation; controlled dataset design.

RESEARCH INTERESTS

- Adversarial machine learning and machine-learning security, including adversarial examples, attack and defence evaluation, threat modelling, and realistic robustness benchmarks.
- Model robustness and reliability under distribution shift, noisy or imbalanced data, calibration error, and uncertainty.
- Graph representation learning for biological networks, link prediction, and heterogeneous data integration.
- Computer vision for anomaly detection, localization, synthetic data generation, and reliable perception in autonomous systems.

LANGUAGE SKILLS AND CERTIFICATION

Mother tongue	Italian
Other language	English - C1 (CEFR)
2022	Cambridge English: Advanced (CAE) Cambridge Assessment English Cambridge English Level 2 Certificate in ESOL International (Advanced), CEFR level C1.

ADDITIONAL INFORMATION

Interests	Classical guitar; cooking; geopolitics.
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